
Current Topics in Digital Philology —

Assignment 3: Historical Linguistics

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1 Exercise 1

Assuming unit costs for insertions, deletions, and substitutions (meaning that they all have a cost of 1), the absolute edit distance between *crocodile* and *cocodrilo* is 3:

1. *crocodile* → *cocodile* (deletion of “r”)
2. *cocodile* → *cocodrile* (insertion of “r”)
3. *cocodrile* → *cocodrilo* (substitution of “e”)

2 Exercise 2

2.1 a

Assuming unit costs for insertions, deletions, and substitutions, the absolute edit distance between *abab* and *baba* is 2. This can be achieved by first deleting the first “a”, and then inserting an “a” at the end of the string:

1. “abab” → “bab” (total cost: 1)
2. “bab” → “baba” (total cost: 2)

2.2 b

Assuming unit costs for insertions and deletions, while assuming a cost of $\frac{1}{3}$ for substitutions, the absolute edit distance between *abab* and *baba* is $1\frac{1}{3}$. This score can be achieved by substituting every character in the string, which, under these costs, is cheaper than the insertion and deletion performed in (a):

1. “abab” → “bbab” (total cost: $\frac{1}{3}$)
2. “bbab” → “baab” (total cost: $\frac{2}{3}$)
3. “baab” → “babb” (total cost: 1)
4. “babb” → “baba” (total cost: $1\frac{1}{3}$)

It can thus be concluded that the optimal alignments for (a) and (b) are not the same.

3 Exercise 3

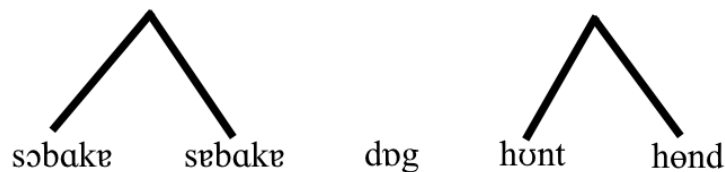
Skipped, as it is not part of the requirements for grade G.

4 Exercise 4

As a technicality, the missing word for "freshwater eel" in Awa Pit might cause issues with an automated cognate judgment program that assumes all cognates have values in all the provided languages. Assuming this is not an issue, I would expect that Cha'palaa and Guambiano words for "saltwater eel" would be marked as cognates, Cha'palaa and Awa Pit words for "shore" are marked as cognates, and that words in all three languages for "water" and "river" are marked as cognates, especially due to the partial cognate for "water" and "river" for Cha'palaa and Awa Pit specifically. Since we've already established these three languages to be part of the same Barbacoan language family, from a linguistic perspective, the ideal outcome would be for the automated cognate judgment to align with our predetermined linguistic classification, that is that words in all three languages would be considered cognates.

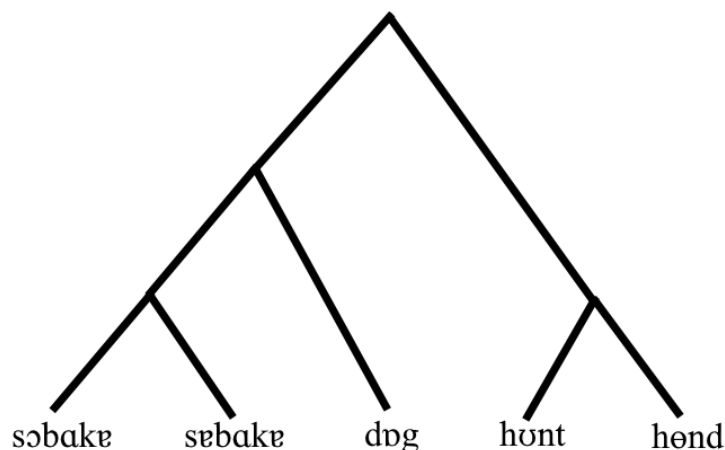
5 Exercise 5

5.1 a



The above figure shows a tree of words where a line between words indicates that they are cognates, assuming a threshold pairwise distance of $t = 0.25$, based on the original pairwise distance matrix. The more closely related two words are, the lower they split in the tree. $t = 0.25$ thus gives three cognate groupings (represented using non-IPA characters): {sobaka, sabaka}, {hunt, hond}, {dog}.

5.2 b



The above figure shows a tree of words where a line between words indicates that they are cognates, assuming a threshold pairwise distance of $t = 0.75$, based on the original pairwise distance matrix. The more closely related two words are, the lower they split in the tree. $t = 0.75$ thus gives one cognate grouping, which contains all the words of the original matrix. Although (non-IPA notation)

“dog-hunt” exceeds the threshold, “hunt-hond” does not, and since the property of cognacy is transitive, “dog-hunt” must also be cognates.

6 Exercise 6

6.1 a

The normalized edit distance between the English and Swedish forms of BITE is the absolute edit distance over the length of the longest form. The absolute edit distance between the English and Swedish forms is 3, and the max length of either of these forms is 5 (“bi:ta”, assuming the vowel lengthening counts as its own character), so, assuming unit costs for insertions, deletions, and substitutions, the normalized edit distance is $\frac{3}{5} = 0.6$. If vowel lengthening is not considered its own character, then the normalized edit distance would be $\frac{3}{4} = 0.75$.

6.2 b

I went with a normalized editing threshold of 0.6 and interpreted the results impressionistically. The automated cognate judgments seem fairly accurate, often marking English and Swedish forms of words as cognates when they are also marked so by experts, while at the same time not marking the word as a Latin cognate when they are not. The program does sometimes struggle when a cognate is shared among all languages. In these cases, Latin is excluded as a cognate when it shouldn't be, likely because the words don't look similar enough.

6.3 c

I impressionistically analyzed changes in automated cognate judgments when changing the threshold from 0.6 to both 0.5 and 0.7. Between these two, 0.6 seems to work best: 0.5 is too strict and produces more false negative cognates than 0.6 does, while 0.7 is too loose and starts to create more false positive cognates than 0.6. I also tried changing the substitution cost to various values, and a cost of 0.8 seemed to give results that might be better than the default unit cost (if only impressionistically).

6.4 d

Both Swedish and English show a palatalization of velar consonants: that is to say that in both languages, words that used to start with a “k” or a “g” sound had their sound changed to become softer, such as (non-IPA characters) “tsh”/“tsj”. An example of this are the forms for SING, which starts with a hard “k” in Latin, but with a soft “s” in English and the Swedish sj-sound in Swedish.

6.5 e

Using a threshold of 0.6 and a substitution cost of 0.8, I received the following results:

- English-Swedish: 0.255
- English-Latin: 0.075
- Swedish-Latin: 0.105

I would say that these numbers match my intuition to how closely these languages are related decently well, although I do find the overall number of cognates low overall. English and Swedish are both Germanic languages, while Latin is both an Romance language and a language that has not evolved in a long time, so I am not surprised to see that English and Swedish are much closer to each other than they are to Latin.